DIPLOMA PROGRAMME PRACTICE · HL · SET B

Chemistry / Paper 2

Worked mark scheme · 90 marks · 150 minutes

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 2 · QUESTION 1 [15]

Locating a period-3 element

(a) Identify X and use the pattern, rather than a lookup, to explain your assignment. [3]

1 The largest jump is between IE_3 and IE_4.

1 Three outer electrons can be removed before a core electron is reached.

1 Period 3 and three outer electrons identify aluminium.

(b) Give the full configuration of X, that of X^{3+}, and the type of orbital from which the fourth electron is removed. [3]

1 X: $1s^2 2s^2 2p^6 3s^2 3p^1$.

1 X^{3+}: $1s^2 2s^2 2p^6$.

1 The fourth electron is removed from a 2p orbital.

(c) Calculate the longest wavelength that can ionize the neutral atom in its ground state. [3]

1 Energy per atom=$578000/(6.02\times10^{23})=9.60\times10^{-19}$ J.

1 $\lambda=hc/E$.

1 $\lambda=2.07\times10^{-7}$ m, or 207 nm.

(d) Compare ionizing a ground-state atom with producing an emission line between two bound states. Relate each to the convergence limit. [3]

1 Ionization removes the electron to the continuum; a bound-state transition leaves it on the atom.

1 Emission releases a photon equal in energy to the difference between the two bound levels.

1 Bound levels become closer at high energy; convergence to the continuum from the ground level corresponds to the first ionization energy.

(e) A 220 nm photon strikes ground-state X. Determine whether single-photon ionization is possible and explain why the atom need not absorb this photon even if its energy is below the threshold. [3]

1 220 nm is longer than 207 nm, so its energy is below threshold.

1 It cannot ionize ground-state X by single-photon absorption.

1 Below threshold a photon must match an allowed bound-state energy gap to be absorbed.

PAPER 2 · QUESTION 2 [15]

Two fluorides in a sealed vessel

(a) Deduce the number of electron domains, lone pairs and molecular geometry around Xe in XeF_4. [3]

1 Four bonding pairs plus two lone pairs give six domains.

1 The two lone pairs occupy opposite positions in an octahedral domain arrangement.

1 The atoms form a square planar molecule.

(b) Deduce the geometry of SF_4 and explain the preferred position of its lone pair. [3]

1 Four bonds and one lone pair give five domains (trigonal bipyramidal).

1 An equatorial lone pair has fewer 90° interactions than an axial lone pair.

1 The resulting molecular geometry is seesaw.

(c) Compare the expected permanent dipole moments and distinguish bond from molecular polarity. [3]

1 The electronegativity differences make the individual bonds polar.

1 Equal Xe–F bond dipoles cancel in square planar XeF_4.

1 The seesaw arrangement in SF_4 does not cancel the bond dipoles; it is polar.

(d) For the Lewis model of XeF_4, show the formal charge on Xe and one F. State why zero formal charge does not imply a nonpolar bond. [3]

1 Xe: $8-4-4=0$.

1 Each F: $7-6-1=0$.

1 Formal charge divides bonding electrons equally by convention; electron density is not shared equally in a polar bond.

(e) Replace one F of square planar XeF_4 by Cl without changing the arrangement. Predict whether the four ligand positions were equivalent before replacement, whether a dipole is expected after replacement, and why. [3]

1 All four positions in XeF_4 are symmetry-equivalent.

1 Replacement preserves cancellation of one opposing F–Xe–F pair.

1 Xe–Cl and the opposing Xe–F do not generally have equal bond dipoles, giving a net dipole.

PAPER 2 · QUESTION 3 [15]

Dissociation of nitrogen dioxide

(a) Explain the sign of ΔS° and determine the standard-state crossover temperature. [3]

1 The number of gas particles increases from two to three, increasing dispersal.

1 At crossover $\Delta G^\circ = 0$ and $T = \Delta H^\circ / \Delta S^\circ$.

1 $T = 114200 / 146.4 = 780$ K.

(b) Calculate ΔG° and K at 700 K. [3]

1 $\Delta G^\circ = 114.2 - 700(0.1464) = +11.72$ kJ.

1 $\ln K = -11720 / (8.31 \times 700) = -2.015$.

1 $K \approx 0.133$.

(c) At equilibrium $p(NO)/p(NO_2) = 0.500$. Deduce the oxygen partial pressure, showing the expression for K. [3]

1 $K = [a(NO)/a(NO_2)]^2 \, a(O_2)$.

1 $0.133 = (0.500)^2 \times p(O_2)/(1.00 \text{ bar})$.

1 $p(O_2) \approx 0.533$ bar.

(d) Oxygen is selectively removed, briefly leaving $p(O_2) = 0.100$ bar and the same $NO:NO_2$ pressure ratio. Calculate ΔG at 700 K and predict the net reaction direction. [3]

1 $Q = (0.500)^2 \times 0.100 = 0.0250$.

1 $\Delta G = 11.72 + 0.00831 \times 700 \times \ln(0.0250) = -9.74$ kJ (approximately).

1 Net forward reaction replaces some removed oxygen.

(e) Determine the heat released by forming 0.750 mol NO_2 via the reverse reaction. Explain why a catalyst does not change the heat per mole for fixed reactant and product states. [3]

1 Reverse $\Delta H^\circ = -114.2$ kJ per 2 mol NO_2.

1 $0.750 \times 114.2/2 = 42.8$ kJ released.

1 A catalyst changes activation energies but not the enthalpy difference between states.

PAPER 2 · QUESTION 4 [15]

A buffer with a limited reserve

(a) Calculate the initial pH using the buffer approximation. [3]

1 n(HA)=0.008000 mol and n(A^-)=0.006000 mol.

1 pKa=3.745; pH=pKa+log(n(A^-)/n(HA)).

1 pH=3.62.

(b) 10.00 cm³ of 0.1000 mol dm⁻³ HCl is added. Show the chemical change and calculate the new pH. [3]

1 $A^- + H^+ \rightarrow HA$ consumes 0.001000 mol A^-.

1 New amounts: 0.005000 mol A^- and 0.009000 mol HA.

1 pH=3.745+log(5/9)=3.49.

(c) Determine the further volume of 0.1000 mol dm⁻³ HCl needed to consume the remaining analytical methanoate. Explain why the original buffer formula then fails. [3]

1 A further 0.005000 mol HCl is required.

1 V=0.005000/0.1000=0.05000 dm³ or 50.00 cm³.

1 The added conjugate base has been consumed; its amount cannot be treated as a large independently fixed buffer component.

(d) At this point estimate the pH from the weak acid equilibrium, using a small-dissociation approximation and checking it afterwards. [3]

1 There is 0.01400 mol acid in 0.1300 dm³, C=0.1077 mol dm⁻³.

1 $K_a = x^2/(C-x)$; x≈0.00431 mol dm⁻³ (or approximately 0.00440 since dissociation is about 4%).

1 pH≈2.37; accept 2.36 with the stated approximation.

(e) Evaluate two claims: dilution by a factor of two doubles buffer capacity expressed per litre, and dilution substantially changes the original buffer pH. Distinguish total amount from concentration. [3]

1 Dilution halves the amounts of acid and base per litre, so capacity per litre decreases.

1 For the whole sample the analytical moles available to consume added acid or base remain the same.

1 The ratio of conjugate-base to acid concentrations remains approximately constant, so ideal buffer pH changes little.

PAPER 2 · QUESTION 5 [15]

Testing an elementary-step model

(a) Identify the intermediate and show that summing the forward steps gives the observed overall equation. [3]

1 N_2O_2 is formed then consumed and is absent from the net equation.

1 Add $2NO \rightarrow N_2O_2$ and $N_2O_2 + O_2 \rightarrow 2NO_2$.

1 Cancel N_2O_2 to recover $2NO + O_2 \rightarrow 2NO_2$.

(b) Derive the predicted rate law in terms of reactants only. [3]

1 Slow elementary step: $\text{rate} = k_2[N_2O_2][O_2]$.

1 Fast pre-equilibrium: $[N_2O_2] = K_1[NO]^2$.

1 Therefore $\text{rate} = k_2K_1[NO]^2[O_2]$.

(c) Predict the rate change if [NO] is halved and $[O_2]$ tripled at fixed temperature; state overall order and units for the observed k when concentration is in mol dm^{-3}. [3]

1 Rate factor $= (\tfrac{1}{2})^2 \times 3 = 0.75$.

1 Overall order is three.

1 k has units $dm^6\ mol^{-2}\ s^{-1}$.

(d) Estimate the apparent activation energy from the temperature pair. [3]

1 $\ln(2.50) = (Ea/R)(1/300 - 1/320)$.

1 $Ea = 8.31\ \ln(2.50)/(1/300 - 1/320)$.

1 $Ea = 36.6\ kJ\ mol^{-1}$.

(e) Assess whether agreement with this rate law proves the mechanism. Suggest additional evidence and explain why the apparent activation energy need not equal the slow step's barrier. [3]

1 Different mechanisms can yield the same empirical law.

1 Detection of transient N_2O_2, for example spectroscopically, would test the intermediate prediction.

1 Observed $k = k_2K_1$, so temperature dependence contains both slow-step and pre-equilibrium contributions.

PAPER 2 · QUESTION 6 [15]

An ester and its hydrolysis products

(a) Use the splitting and integration of the doublet and septet to identify a fragment and its location relative to oxygen. [3]

1 A 6H doublet indicates two equivalent methyl groups adjacent to one proton.

1 A 1H septet indicates a CH coupled to six equivalent methyl protons.

1 The downfield CH is bonded to ester oxygen: $–OCH(CH_3)_2$.

(b) Deduce the complete ester structure and explain the singlet, giving the systematic name. [3]

1 Structure $CH_3C(=O)OCH(CH_3)_2$.

1 The acetyl methyl has no adjacent proton across the carbonyl carbon, so is a singlet.

1 Propan-2-yl ethanoate (isopropyl acetate accepted).

(c) Write the equation for alkaline hydrolysis and calculate the minimum volume of 0.2500 mol dm^{-3} NaOH needed. [3]

1 $CH_3COOCH(CH_3)_2+NaOH→CH_3COONa+(CH_3)_2CHOH$.

1 n(ester)=2.040/102.0=0.02000 mol; one equivalent hydroxide is needed.

1 Volume=0.02000/0.2500=0.08000 dm^3 or 80.00 cm^3.

(d) An excess of aqueous acid is then added to the cooled mixture. Identify the organic acid formed, write the responsible ionic equation and identify a diagnostic IR change for the isolated dry acid compared with the original ester. [3]

1 Ethanoic acid forms.

1 $CH_3COO^-+H^+→CH_3COOH$.

1 The acid has a very broad O–H absorption around 2500–3300 cm^{-1}, absent in the ester.

(e) Predict the main product on controlled oxidation of the alcohol hydrolysis product and identify two expected changes in its ^1H NMR spectrum. [3]

1 Propan-2-ol gives propanone.

1 The proton on the carbon bearing OH is absent in propanone.

1 The methyl doublet becomes one 6H singlet since there is no neighbouring CH proton.